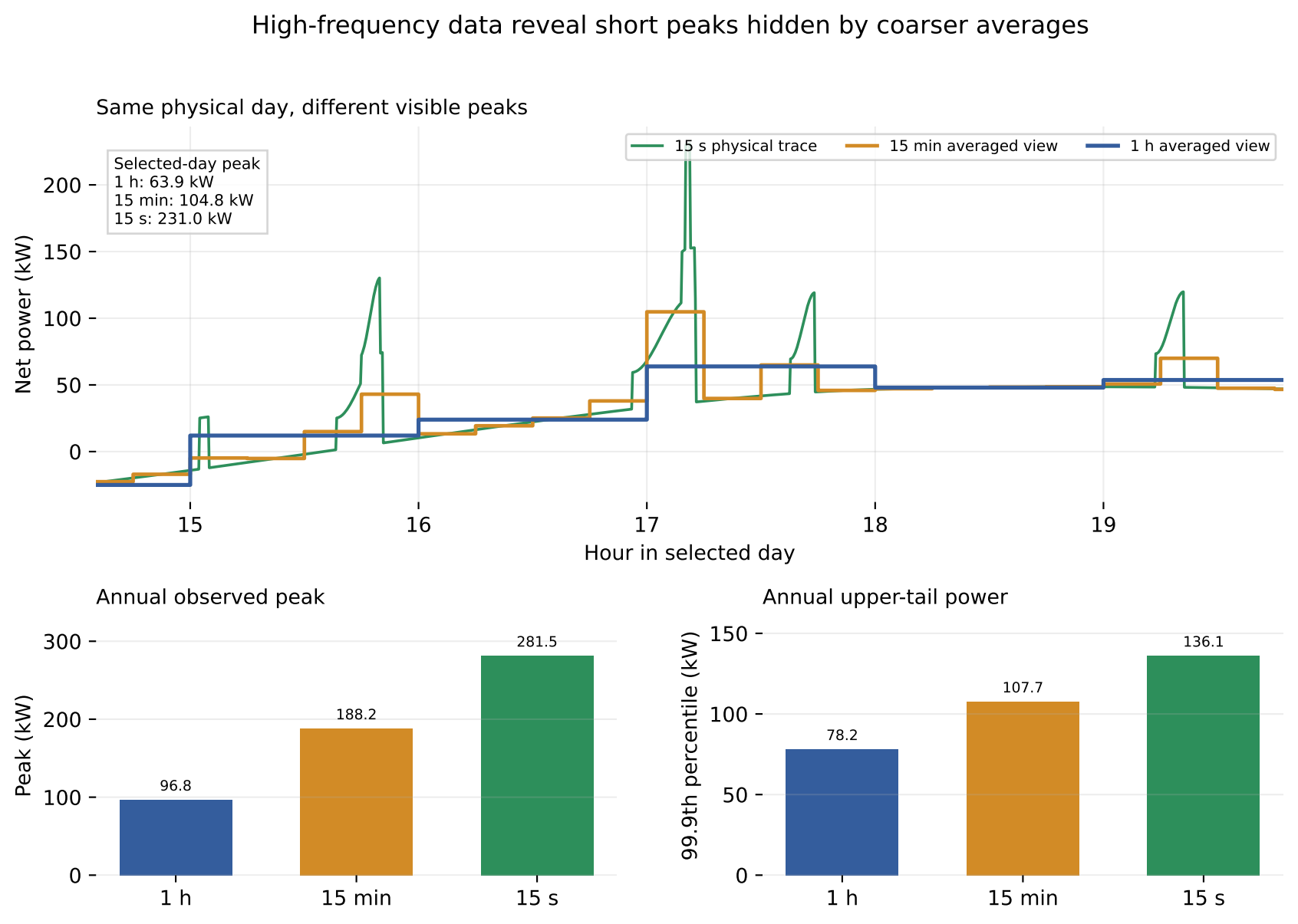
High-frequency data reveal short peaks hidden by coarser averages
Same physical day, different visible peaks
15 s physical trace
15 min averaged view
1 h averaged view
Selected-day peak
1 h: 63.9 kW
15 min: 104.8 kW
15 s: 231.0 kW
Net power (kW)
200
150
100
50
0
15
16
17
18
19
Hour in selected day
Annual observed peak
Peak (kW)
300
200
100
0
96.8
188.2
281.5
1 h
15 min
15 s
Annual upper-tail power
99.9th percentile (kW)
150
100
50
0
78.2
107.7
136.1
1 h
15 min
15 s